

Hello!

I am Sayeed Unnisa

I am here because I love to give presentations on **Java Full Stack**

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The background is a solid teal color with a pattern of thin, wavy, concentric lines that create a sense of depth and movement. In the upper right and lower left corners, there is a subtle pattern of small, light-colored crosses or plus signs arranged in a grid-like fashion.

JSP

JSP

- JSP stands for Java Server Pages.
- It is a server side technology used for creating Dynamic web applications
- A JSP page consists of HTML tags and JSP tags.
- In this, JSP tags are used to insert JAVA code into HTML pages.
- It is an advanced version of Servlet Technology.
- JSP is first converted into servlet by JSP container before processing the client's request.

JSP pages are more advantageous than Servlet:

They are easy to maintain.

No recompilation or redeployment is required.

It has all features of servlets and in addition to this it has implicit objects and custom tags

Jsp Translation:

each jsp page will be translated into a servlet in server

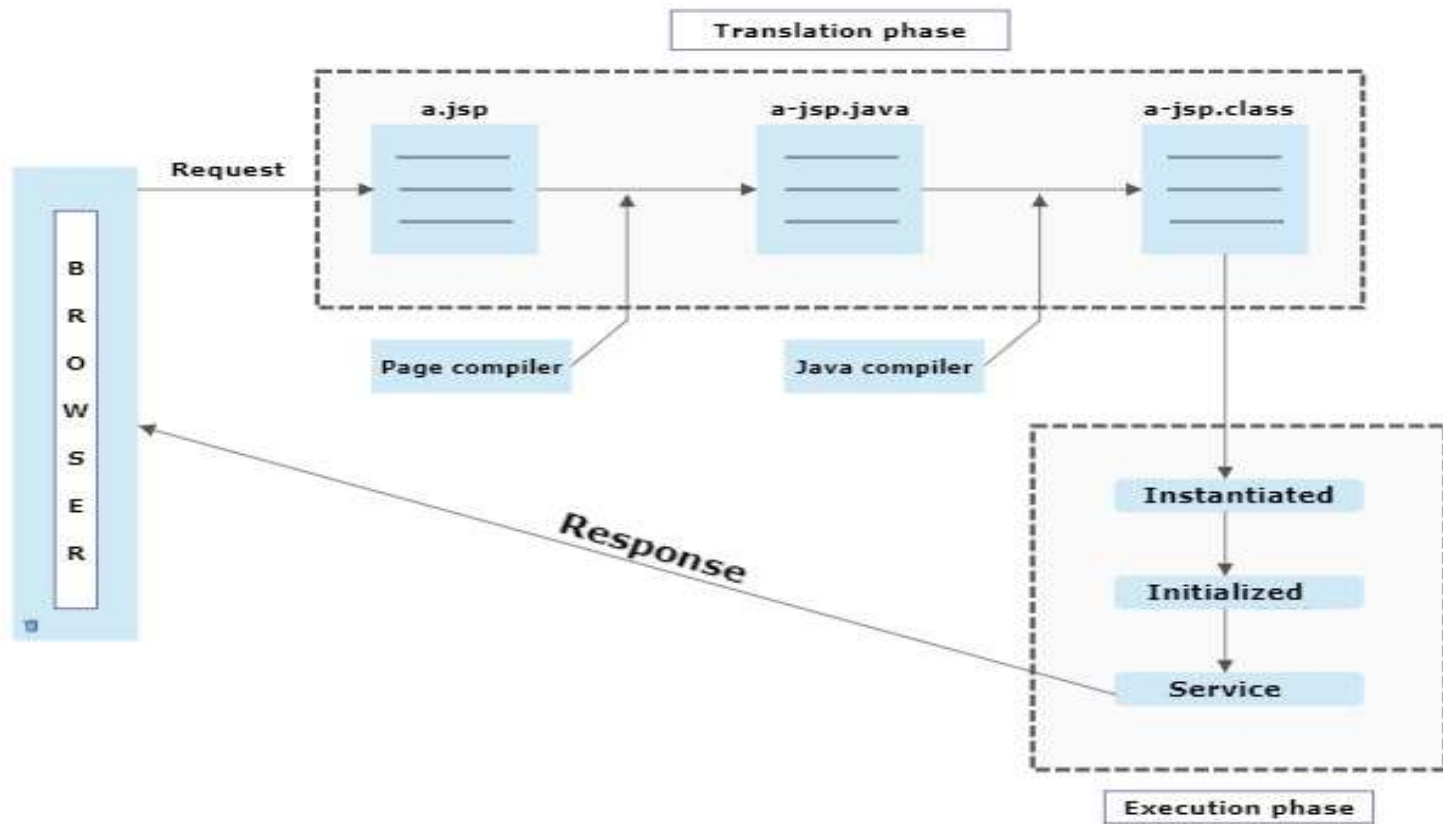
A jsp execution in server has two phases

- **Translation phase**

In translation phase, first a jsp page will be converted to equivalent servlet and then a servlet will be compiled.

- **Execution phase**

In execution phase, object is created for a servlet then it is initialized then finally service() is executed



JSP CODE

In JSP File, we should write everything as Tag format only.
We can write HTML Tags and JSP Tags.

Types of JSP Tags:

1.Scripting Tags:

A programming code written in a page is called script.

In order to write java code in jsp pages, jsp technology has provided scripting tags

scripting tags are 3 types.

1. scriptlet

2. expression

3. declaration

2. Directive Tags:

It is used to Link other sources with this file. Like import statements, link with other JSP, Session linking, error pages etc.

3.Action Tags:

It is used to perform operations over data and flow.

Scriptlet:

This scripting tag is used for writing some java code in a jsp, which we want to execute

Syntax:

```
<%  
    // java code  
%>
```

- At the time of translation, page compiler inserts all the scriptlet tags code into **_jspService()** method
- In a scriptlet tag, we can create variables, but we cannot create methods because in java we cannot define one method in another method.
- The variables created in scriptlet tag will become local variables to **_jspService()** method

Expression:

This scripting tag is used for defining an expression in a jsp page at run time the expression is executed and its result will be printed on browser for each request

Syntax:

```
<%  
= expression  
%>
```

- At the translation time, page compiler converts expression tag into `out.print()` statement and inserts that statement into `_jspService()` method

Declaration:

This tag is used to create global variables and methods in a jsp page

Syntax:

```
<% !  
//global variables  
//methods  
%>
```

The code defined in declaration tag will be inserted into class and the code defined in scriptlet and expression tags goes to `_jspService()`.

Note:

Jsp comment

Ex : `<%--this is jsp Comment --%>`

2.Directives:

Directives are used to give direction to the page compiler

These tags are used to link other resources (like import, include etc).

Syntax:-

`<%@ Directive_name attribute="value"%>`

Types of directives:

- page Directive
- include Directive

i. include:

This directive tells page compiler to include the source code of another page into the current jsp

include directive has only one attribute called "file"

Ex:

```
<%@include file="b.jsp"%>
```

ii. page:

The page directive defines attributes that apply to an entire JSP page.

Syntax

```
<%@ page attribute="value" %>
```

Attributes of JSP page directive

- **import** : import attribute is used for importing java packages in jsp
- **session** : This attribute is used to either enable or disable session.
- **isELIgnored**: This attribute is used to either enable or disable Expression Language(EL) (default value is false).
Expression Language is enabled by default

3.Action tags:

These tags are used to avoid “writing of JAVA code in JSP”.

We can also called as pre-defined tags in JSP.

jsp can communicate with another jsp or servlet or html or java bean

The action tags:

- 1.<jsp:forward>
- 2.<jsp:include>

1.<jsp:forward>

- this action tag is used to forward a request from a jsp page to another jsp page or servlet or html
- this action tag is similar to forward() method of RequestDispatcher in servlet
- when forwarding a request, if we want to forward any parameters then use <jsp:param> sub tag under <jsp:forward> tag

Syntax:-

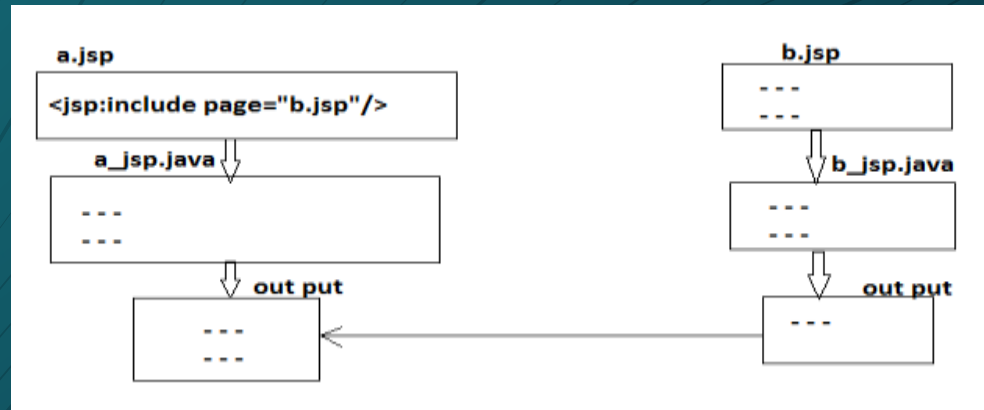
```
<jsp : forward page="-----.jsp"/>
```

Ex:

```
<jsp:forward page="b.jsp">  
<jsp:param name="p1" value="100"/>  
</jsp:forward>
```

2. <jsp:include>

- This action tag includes output of another page into output of current page.
- This action tag is similar to include() method of RequestDispatcher of servlet
- In jsp we have include directive and include action tag
- Include directive is going to include source code & include action is going to include the output



Ex:

```
<jsp:include page="b.jsp">  
<jsp:param name="p1" value="100"/>  
</jsp:include>
```

scopes in jsp:

Scope decides the accessibility of an object or variable.

JSP provides 4 scopes to a variable.

1. Page Scope.
2. Request Scope.
3. Session Scope.
4. Application Scope.

Page Scope:

Page Scope makes variable available to the current page only where it is defined.

Request Scope

- Request Scope makes variable available to the current request only
- A JSP object created using the 'request' scope can be accessed from any pages that serves that request.
- Single request may include multiple pages using forward.

Session scope

- Session is a particular interval of time in which communication between the client and the server is happening with HTTP.
- 'session' scope means, the JSP object is accessible from pages that belong to the same session from where it was created.

Application Scope

- A JSP object created using the 'application' scope can be accessed from any pages across the application.

implicit objects:

JSP Implicit objects are created by the web container and these are available to every JSP page of application

There are 9 implicit objects available in the web container

1. out
2. request
3. response
4. config
5. session
6. application
7. page
8. pageContext
9. exception

Out:

An instance of the JspWriter class that allows you to write content to the output stream that will be sent to the client's browser.

request:

Represents the client's request data to the server and provides methods to access parameters and attributes associated with the request.

response:

Represents the server's response to the client and provides methods to control the response

Session:

It is used to get session information.

Represents the user session and provides methods to manage session attributes.

Config:

It is used to get the information regarding configuration of JSP pages.
For example get Servlet Name etc.

Application:

This object is created by the web container only when an application is deployed.

It can get set or even remove the attributes under application scope

pageContext:

This object holds all page related data and other objects internally. This object can store other objects data like page, request, session and application scopes.

page:

"page" object is same as "this" object.

while creating a servlet of jsp page compiler assigns "this" in to "page"

Exception:

Exception is an implicit built-in object of java.lang.Throwable. This is useful for handling exceptions in JSP.

attributes in jsp:

- Attributes are used for sharing the data.
- A jsp can share its data with other pages through attributes
- In order to share the data a scope need to be selected.
- In jsp attributes can be stored either in page or request or session or application scope.

Ex:

```
pageContext.setAttribute("k1","java",PageContext.PAGE_SCOPE);  
pageContext.setAttribute("k3","servlet",pageContext.SESSION_SCOPE);
```

Expression Language:

- Expression Language is mainly used to eliminate java code from the JSP
- The main purpose of it to simplify the process of accessing data from **bean** properties and from **implicit objects**.

Syntax of EL:

`${expression}`

whatever present inside braces gets evaluated at runtime and being sent to the output stream.

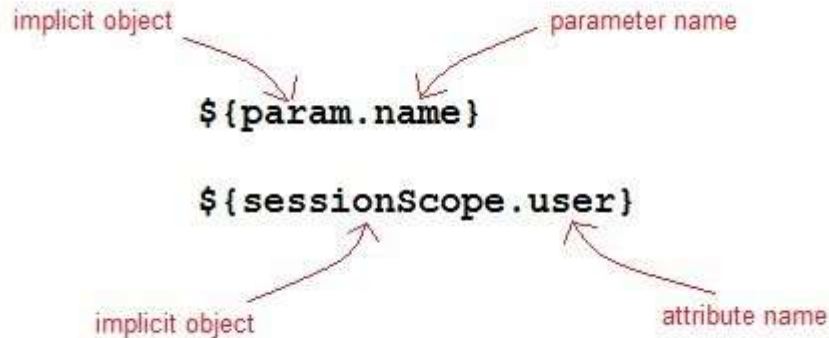
Example

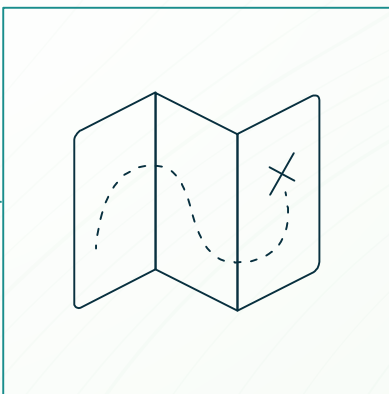
To print the value of request parameter uname:

```
<% uname=request.getParameter("uname");%>
```

by using expression language syntax.

```
${param.uname}
```





Thanks!

Any questions?

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